

Performance Evaluation of Space Data Centers: DDSSO-Assisted LEO Mega-Constellations

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ABSTRACT

Space data centers (SDCs) have been proposed as an alternative to terrestrial data centers, which face critical challenges in power consumption, water usage, resource waste, and carbon emissions. In particular, the dawn-dusk sun-synchronous orbit (DDSSO) assisted low earth orbit (LEO) mega-constellation, so called DDSSO-LEO, has gained increasing attention. However, the performance of DDSSO LEO has not been closely evaluated. This study simulates a DDSSO-LEO network and evaluates its performance in terms of latency and task success rate (TSR). We further analyze how the performance gain varies with DDSSO computing capability, orbital altitude, and task size, identifying when DDSSO-LEO SDC is beneficial compared to LEO-only SDC.

Proposal of Space Data Center Architecture

Introduction

The rapid growth of AI applications and data demand has increased the importance of data centers, while terrestrial data centers face critical issues such as excessive power consumption, water scarcity, and resource depletion. As an alternative, space data centers have been proposed by deploying computing resources on satellites. Google and SpaceX both suggested dawn-dusk sun synchronous orbit (DDSSO) as a promising orbit for this purpose. Since DDSSO enables stable solar power generation, a space data center can be realized as a DDSSO-assisted low earth orbit (LEO) mega-constellation, DDSSO-LEO, where DDSSOs provide high-performance computing and LEO mega constellations support global connectivity with low-performance computing.

Proposed Network Architecture

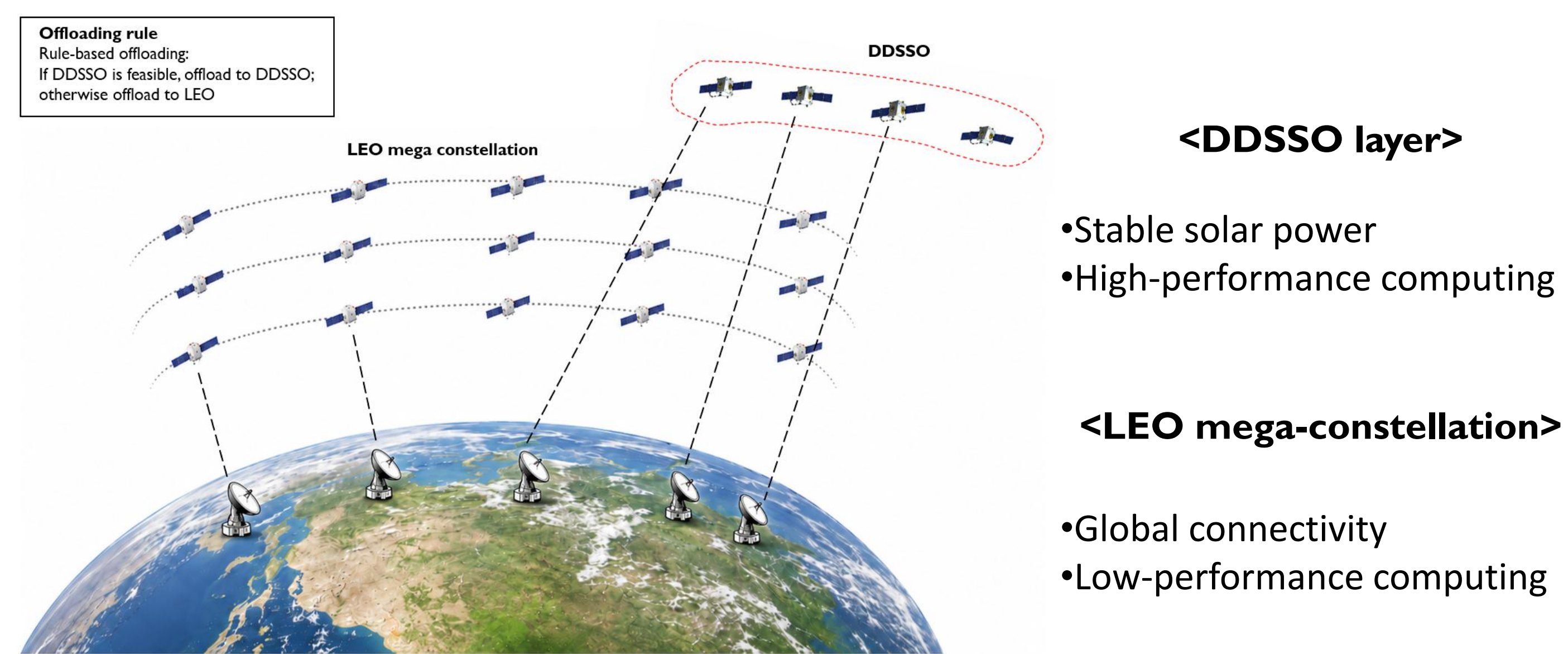


Fig. 3. Schematic illustration of the SSO

Sun-Synchronous Orbit

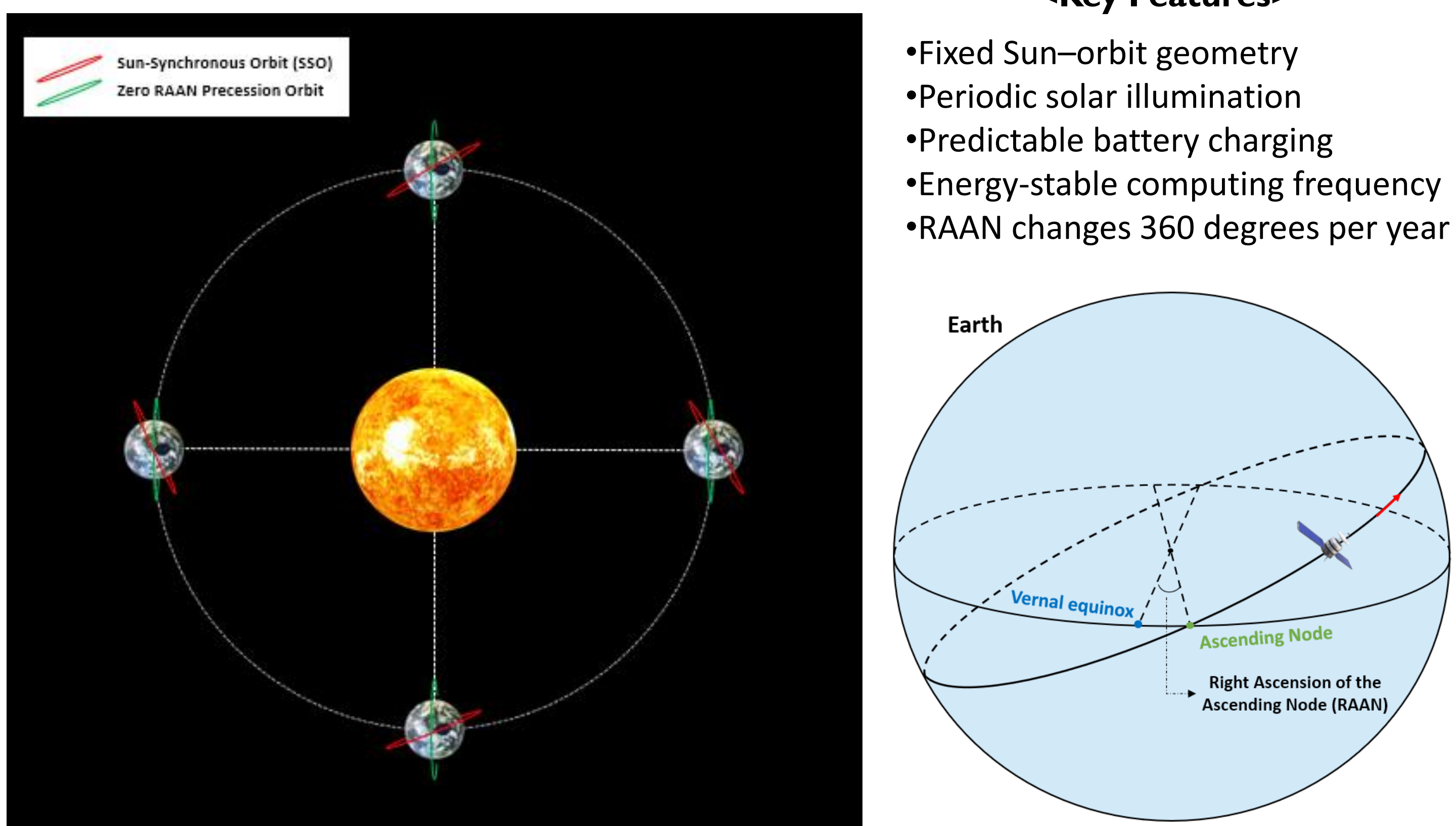


Fig. 3. Schematic illustration of the SSO

Fig. 4. RAAN definition illustration

Simulation results

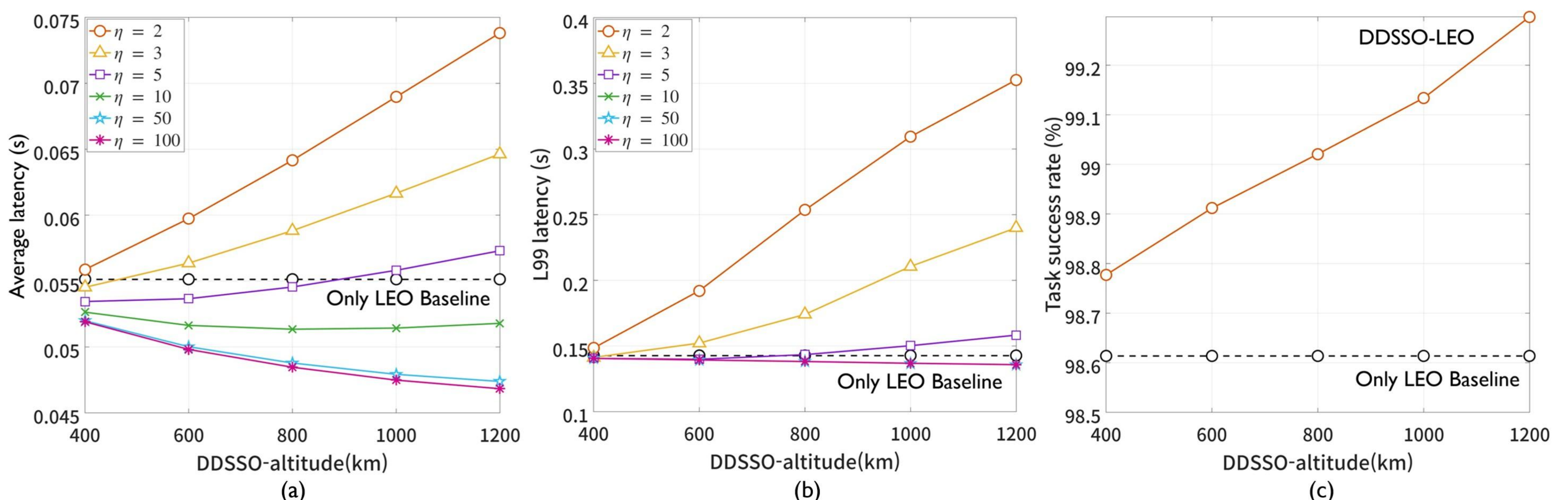


Fig. 5. The Sun-Synchronous Orbit-assisted LEO Mega-Constellation

System model

A. Feasible set

We set a whole set of LEO satellites (LEOS) and DDSSO satellites (DDSSOS) as follows, $V_L = \{v_1, v_2, \dots, v_L\}$, $V_D = \{v_1, v_2, \dots, v_D\}$. Each gateway k can access only a subset of LEOS and DDSSOS whose elevation angle lies in $25^\circ \leq \theta_{k,v}^\psi \leq 90^\circ$.

Thus, we define each feasible set of LEOS and DDSSOS as follows, respectively,

$$V_{k,L,c}^\psi = \{v_l | v_l \in V_L, 25^\circ \leq \theta_{k,v_l}^\psi \leq 90^\circ\} \quad (1)$$

and

$$V_{k,D,c}^\psi = \{v_d | v_d \in V_D, 25^\circ \leq \theta_{k,v_d}^\psi \leq 90^\circ\} \quad (2)$$

B. Computing offloading model

For a task generated by gateway k at slot ψ , we define the binary offloading variables as $a_{k,v}^\psi, b_{k,v}^\psi$. Each variable is 1 if the gateway k offloads to LEOS and DDSSOS respectively, and otherwise 0.

In this paper, we consider a naive rule-based offloading policy where gateway k offloads DDSSOS whenever DDSSOS is feasible. Also, we define N_{task} as the total number of generated tasks and S as the set of successfully served tasks.

C. Computing frequency

The computing frequency is defined as below,

$$f_{D,c,max} = \eta \cdot f_{L,c,max}, \quad (3)$$

where $f_{L,c,max}$ and $f_{D,c,max}$ represent the maximum computing frequency of LEOS and DDSSOS, respectively, and $\eta \geq 1$ denotes the computing frequency ratio.

D. Evaluation metrics

To evaluate the QoS of the proposed network, we define the latency of the task generated by gateway k at slot ψ as

$$t_k^\psi = t_{r,k}^\psi + t_{p,k}^\psi + t_{c,k}^\psi, \quad (4)$$

where $t_{r,k}^\psi$, $t_{p,k}^\psi$, and $t_{c,k}^\psi$ denote the transmission delay, propagation delay, and computing latency, respectively.

We evaluate the performance using three metrics: average latency, L99 latency, and task success rate (TSR). The average latency is defined as

$$T_{\text{avg}} = \frac{1}{|S|} \sum_{(k,\psi) \in S} t_k^\psi. \quad (5)$$

The L99 latency is defined as the 99th percentile of the latency values, i.e.,

$$T_{L99} = \text{Percentile}_{99} \left(\{t_k^\psi\}_{(k,\psi) \in S} \right). \quad (6)$$

The TSR is defined as the ratio of the number of successfully served tasks to the total number of generated tasks:

$$P_{\text{tsr}} = \frac{|S|}{N_{\text{task}}}. \quad (7)$$

Limitations and Future Work

- This study provides an underlying performance evaluation of a DDSSO-LEO under a naive offloading policy.
- To further improve network performance, **software-defined networks (SDN) should be introduced** into the multi-layer DDSSO-LEO architecture.
- **Through SDN-based control, task offloading, path selection, and resource allocation can be jointly optimized according to task size or service type.**
- Since both the LEO and DDSSO layers are highly dynamic, an SDN controller should support topology-aware path selection and resource allocation across heterogeneous orbital layers.
- Practical deployment issues such as the launch cost of DDSSOs, altitude-dependent radiation effects, and thermal management should be jointly investigated in future work.

CONCLUSION

In this study, we evaluated the underlying performance of a DDSSO-assisted LEO mega-constellation, DDSSO-LEO, in terms of latency and task success rate. The simulation results demonstrate that **DDSSO computing frequency, DDSSO altitude, and task size** are key factors that affect the network performance of the DDSSO-LEO. DDSSOs can outperform the LEO baseline only when their computing frequency is sufficiently high to compensate for the additional propagation delay. Thus, an appropriate computing frequency and orbital altitude should be jointly selected according to the target service. In addition, the offloading decision between LEOS and DDSSO should be determined according to the service requirement, rather than using a fixed-rule based policy.